

ActInf ModelStream 020.1: Kenric Nelson: "Learning extreme models with

ModelStream | 1:10:33

Hello, welcome everyone. This is Active Inference Model Stream number 20.1 on January 30th, 2026. And we will be discussing learning extreme models with a coupled free energy

by Kenrick Nelson here and colleagues. So Kenrick, thanks a lot for joining.

You will provide a presentation. And if anyone is watching live, feel free to put any questions in the live chat and I will read them when we have a discussion. So thanks again for joining and to you for the presentation. Thank you, Daniel. Really appreciate you hosting this talk.

Folktrek has been doing a deep dive into Active Inference, including reading the new book that's come out in the last few years. And so we're looking forward to talking with your community about our own work in modeling extremes and how that can be used to improve artificial intelligence.

My colleagues that worked on this presentation with me are Igor Olivera from Brazil and Amina Al-Najafi from Iraq. And so what I'm going to go through is sort of an introduction to complex systems and why they have properties that are unique in terms of trying to do modeling in AI.

And then I'm going to go through a variety of entropy measures, including the standard ones of the Shannon entropy, but also generalizations that apply to complex systems. And then show why there's a unique one that I call the coupled entropy and how we can use that in variational inference to generalize the free energy function and thereby be able to model quite extreme environments and their distributions.

And we'll go over some machine learning about results for that. And then I look forward to discussing it with you and the community. So FOTREC is building capabilities and risk aware intelligence.

There's a couple aspects of that. There's a technical aspect, which I'm going to brief today with regard to our work in machine learning and risk risk. There's also a Dow governance piece in terms of using ideas from complex systems regarding how you design protocols for our communities to make good decisions. And then we see these two threads coming together in terms of how we can better mitigate extreme risks like the issues around climate change. The team is diverse. I'm in the Cambridge, Massachusetts area, but we have working partners around the globe. And then we also collaborate with universities. I was on the faculty at Boston University, collaborating with Mark Khan for several years and Eugene Stanley. And then I have also collaborated with Sabir Umaroff at the University of New Haven. And then we have several other university partners. I was in Turkey for two months this year collaborating with Ugar Turknali, which was a fantastic opportunity. So here's the challenge. When it comes to

universities. I was on the faculty at Boston University, collaborating with Mark Khan for several

years and Eugene Stanley. And then I have also collaborated with Sabir Umaroff at the University of

New Haven. And then we have several other university partners. I was in Turkey for two months this year collaborating with Ugar Turknali, which was a fantastic opportunity. So here's the challenge. When it comes to

catastrophic risk, we know that it's that those very low probability events that have high impact that are a real concern. But machine learning has grown out of traditions in statistics that revolve around what's called the exponential family and the Gaussian being the most popular example of that.

And even the metrics are based on the inverse of that, which is the logarithmic function. That gives us the maximum likelihood metric that gives us the Boltzmann gives Shannon entropy. But these distributions and these metrics don't really account for the heavy tailed properties and the nonlinear properties that we

experience in complex systems. And they're quite difficult because the moments can diverge. So there can be

situations where you can't measure the variance, which is the most common thing that you're trying to characterize in machine learning. So we need ways to account for this. And just as one example to visualize this, you know, the cryptocurrency space is one of the areas that we work in as a community. And Bitcoin is notoriously very volatile. And you can see that in these spikes that are not modeled by a Gaussian distribution. Here we have in the middle an agent based model where in red, you would see something like a Gaussian process. And you can see that yes, there's noise, but there's not these large fluctuations. But we can model

these with a distribution that I refer to as the coupled Gaussian. And the coupling has to do with the fact that we can map the tail decay shape to the source of nonlinearity in the system. And I'm going to go into more detail about that.

So we're generalizing the exponential family. And just to remind people what that is, first of all, of course, it evolves around the exponential function. And then you have a set of parameters that are dot multiplied with a set of statistics, the traditional ones being just x and x squared. There can be some other terms that involve x . But we'll typically just set that h to one for today's discussion. And then all of this is normalized by what's called the partition function. When we go to the coupled exponential family, what we're doing is we're modifying this exponential function to be a power law type of function. And I show that here. So this coupled exponential, I'll highlight it right here, whoops, I got ahead of myself.

It multiplies the variable by this coupling, adds one, and then renormalizes, or rather takes it to a power one over the coupling. And that plus one is a critical feature of the map. That's what enables it to converge back to the exponential. But it's also what complicates things in terms of analysis of nonlinear systems. Now we keep the statistic and the parameters. And there's an additional factor here.

This one plus d kappa has to do with the change from a cumulative distribution. When you take the derivative, you get this additional parameter in the exponent. The most common distributions that we'll talk about today are the generalized Pareto distribution, which I refer to as the coupled exponential. That's when this alpha term is equal to one. And the student t distribution, when alpha is equal to two. And I refer to that as a coupled Gaussian. The critical thing is that the moments diverge.

And this, as this coupling grows, more and more of the moments diverge. And I'll go into more detail of that as we go forward. But taking a look at an example here, that's going to be used a lot in machine learning, because this is the distribution that will, you know, be our latent distribution, is the coupled Gaussian. So in black here, you'll see the traditional bell curve with the coupling equal to zero, that's your Gaussian. And that's what traditional machine learning algorithms are built around.

In blue is what's called a compact support distribution, where it actually decays faster and truncates. I'm not going to talk a lot about that today, but that would be a negative coupling.

And then what we're really focused on is these heavy tail distributions. In this case, the one in red has a coupling of one, which is the Cauchy distribution. And the critical insight in defining this family is that the shape, that tail shape parameter, is exactly equal to the nonlinear

statistical coupling, or the source of nonlinearity in the system. And these are proportional to a measure of complexity of the system. So we're going to show that we can model now the most extreme complex systems with these techniques. One of the early insights about this work that I showed in a paper several years ago, that actually introduced the coupled entropy function, was a concept of an average density. So it's, people are fairly familiar that the center of a distribution is the mean, and that the variation is expressed for the Gaussian as the standard deviation. And more generally for these non-exponential distributions, it's referred to as a scale. But people have a difficult time conceptualizing what entropy is measuring. You should think of entropy as the measurement of uncertainty along the y-axis of a distribution. And you can always take the entropy, which is in the log space, take the inverse of that, the exponential, and bring it back to the probability space. When you do that, you get a function that I refer to as the average density of the distribution. And this is a translation of a generalized entropy function. And what the key thing here is that that average density always lines up at the scale. This is the invariant. And so this particular scale of the distribution has a number of things that are unique about it. And in particular, what I've proven this past year is that while there's literally dozens of candidates for generalizing the Boltzmann-Gibbs-Shannon entropy, there has, during this 50-year period of development in the modeling of complex systems, been a missing requirement. And that requirement is that your generalized entropy has to be a measure of this particular point of the distribution where you have what I call the information scale. And the reason that's important is because that point in the distribution is sensitive to the linear sources of uncertainty, whereas the tail decay is sensitive to the nonlinear sources of uncertainty. And these two parameters are the only way to precisely separate the role of linear and nonlinear processes. So let's take a closer look. These distributions here are from the family of the coupled exponentials or the generalized Pareto distribution. So if we start at the top, this is one of those compact support distributions. And we can see it for two different parameters of the scale. On the top, the scale is equal to two. And on the bottom, the scale is equal to half. And you can see here clearly how it truncates. Now, the Boltzmann-Gibbs-Shannon is, in that case, lower than this scale point. And when we go to Heavytail, these lower curves, what happens is the Boltzmann-Gibbs-Shannon scale gets, its probability gets very low. And that is to say, its entropy gets very high. And it becomes dominated by the tail shape. And you lose sensitivity to this scale parameter. So this is what motivates the need for a generalization. And the first significant generalization was done in the 1950s and 60s by Alfred Renyi, where he showed that you could match a generalized mean to that tail shape and then still take the natural logarithm and get a very effective measurement of uncertainty in these complex systems. And indeed, it's a good metric. It's used quite a lot in machine learning and in other applications. But it doesn't match up with this, what I'm saying is a requirement here to actually measure the shape, the uncertainty right at this scale. So there was another significant proposal that dates back to 1988 and has gotten a lot of attention recently known as the solace entropy. It introduces a concept that I'm going to talk about more that's

quite important called an independent equals distribution. However, I've since proven that it is not in fact the correct function for generalizing entropy. And there's a missing piece to it that was left out. But if you look at this requirement, this is what I'll show is equal to this coupled entropy that I've defined. I'm going to show that in a moment. But the fascinating thing about this is that the solution right along this scale is equal to one plus a generalized logarithm of the scale. And this is the beauty of it. And this was sort of the eureka moment when I saw this solution that I realized that there is something quite unique about the coupled entropy and about this requirement. Because now I'm maintaining in my entropy function a dominant metric about this scale parameter. And the only thing I'm doing is I'm changing what logarithm function is measuring that. So let's dig into it a little bit more and see what is where this function comes from. So as I said, we start with a coupled exponential function that forms these non-exponential family of distributions. And then we take the inverse of that to form the metrics known as the coupled logarithm. And here you can see clearly that the coupling is directly that power. And there's a minus one in order for it to converge properly. And there's a normalization. Now, when we go to forming the entropy function, there's some additional features that have to come into this. So the first thing you'll see is that this generalized logarithm now is the coupling times alpha. You can think of alpha as the shape near the location of the distribution. So again, the most common ones is one for the exponential distributions, two for the Gaussians, but it can be a real number. So the other critical piece, and this is what was left out of the formation of the solace entropy, is that you can't just take this natural, this generalized logarithm of the probabilities. You have to account for how dimensions affect the metric. And this, again, has to do with the fact that you can think of the coupled logarithm and the coupled exponential being appropriate that you can't do with the cumulative distributions. But once you take a derivative, you have to account for additional terms in the exponent. And each dimension involves another derivative. So you get this factor here that you have to raise the probabilities to. Now, this term in the right, this generalized logarithm, let me go back, that is exactly equal to the inverse of this generalized exponential. And so what that gives you is the interior of this. And you can bring this inside the parentheses, too. I'm not going to show that today. But basically, you're going to get the argument of this distribution. However, that's not enough. There's another piece here that pulls the whole thing together. And that is that you do not average over the distribution itself. You average over something that's I refer to as the independent equals distribution because what this is modeling is how many random variables are sort of independent but equal to each other. And this is the metric that will give you the parameters of that distribution. So this is the proper statistic. That function, I can show here, it's taking the original distribution, p , and raising it to this power and then renormalizing. And the effect of this, particularly all the way here on the right, is that it reduces the original coupling value. So now I have a modified coupling value. And the critical thing is that as you go to infinity, you always, you know, for alpha equal to 2,

is the case we're going to be most interested in, this will converge to a half. And that half is the boundary where the distribution would go to infinite variance. So now what I have is this balanced metric in which I'm going to put a very high penalty on low probability events through this generalized logarithm. But at the same time, I'm going to sample over a distribution that makes sure that those rare events events don't occur too often. So let's see what the impact of this is now on variational inference. Um, if, you know, Dan, if there's been any questions in the chat, this would be a good place to break and maybe answer some questions. Um, let me just ask one clarifying question. What is coupled, what is coupled to what? Yeah. Okay. Excellent question. So I originally coined this phrase nonlinear statistical coupling, because you can think of it as treating the states of a distribution as not being independent from each other, but rather the states are coupled together. Um, it comes out explicitly from this, um, coupled exponential, this, this independent equals distribution. You can kind of rearrange this function and show that it's also equal to multiplying all of the states together where you have the state you're interested in, but the other states are raised to this power. Now, since then, what's become more relevant is the fact that this coupling term is the source of non-linearity in the system. So a classic example, which I'm not going to show today, but is very relevant to the fluctuations in Bitcoin is multiplicative noise. So, um, our, our link, when we studied linear systems, we typically have a deterministic component that we're trying to learn. And then we model, say the Gaussian noise as being additive, but in complex systems, you're going to have a third term in which the deterministic process and the noise process are multiplied together. And you can show that, um, the coefficient on that third term is equal to this coupling. Okay. So just to kind of echo that back in a Gaussian mixture model, whether you have just one Gaussian or multiple Gaussians, you're superimposing different means and variances, but the variances never interact. It's just kind of like you're stacking a bunch of Gaussians on top of each other. Whereas here we can get something that's looking more like multiplicative noise because of this multiple, um, coupling that, that extends the tails of the distributions.

And I guess maybe address it at the end or when, right. But how do we know how much to inflate those tails? When, how do we calibrate that factor to empirical data? And how would that differ in its calibration or, or, or precision or recall than just using one of these other traditional heavy tail distributions?

Yeah. All right. So I, that's a good, I'll answer that now as a preview of where we're going to go with the talk. Um, so one thing we haven't shown all of this, but I would argue that this coupled exponential family has the ability to model any heavy tail distribution, because what I'm doing is that coupling parameter is, um, the asymptotic tail decay. So you can figure out what that is.

And then the parameters and statistics measure our measurements of all, all that's happening in the

distribution in between. So there's not a limitation in what kinds of distributions you can model with this method. That's number one. Number two is a very tricky question. Um, I'm going to talk about the fact that the way I see it, there's kind of two purposes for the coupling, two or three. Um, the first is you're describing is the system itself that you're trying to learn can have these fluctuations, and you're going to want to characterize that and then use that in the way we're doing it right now. It's a hyper parameter, but you can, you can analyze the data and determine a tail that fits to the data you're interested in. That's one. But I think actually the more important thing is a way to make systems robust, because whenever we train, we always have limited data. And we also also have data that might not exactly fit what we know we're going to experience out in the field. So I, we can also use this tail decay as a way to improve the robustness against things that weren't in the training set. So there's these two components to how you will use the tail. All right, with that, I'll press forward. Um, so we're particularly interested in how we can apply this to variational inference and active inference. So variational inference is a Bayesian learning method where we try to approximate a complex true posterior distribution with a tractable distribution where we're going to learn the parameters of that distribution. And the goal is to minimize the divergence between the true posterior and the, and the approximate model. Now recently, and what Daniel is, uh, and his team at the active inference Institute is focused on is that this basic framework has now been applied much more broadly to neuroscience, artificial intelligence, and even, uh, socio-economic problems where you can think of every single node in a neural network as doing a comparison between a top-down prediction or a bottom-up observation. And then the difference becomes a way to, um, build an inference model as you optimize and minimize those differences. Likewise, you can apply this to actions and policies by using the prior or the prior top-down, um, as a preference of what you want to have happen. And then the observations are now expected observations, and then that can be, that can control, uh, the actions of a system. So the challenge is how can we modify the cost function, this evidence lower bound or the negative free energy, uh, to be able to learn heavy tail distributions. Um, so let's, we're in our own experiments today, we've been using the variational auto encoder as a foundation for the experiments. So let me just go over what that is. Um, here you can take a complex dataset such as, uh, the Celeb A, uh, photographs, and you use a neural network to encode a mean and a covariance matrix. Again, in the standard VAE, this is a Gaussian or a mixture of Gaussians that you're modeling in the latent space. And then you can sample from that and decode it in order to get a new sample generated. And the thing that you're optimizing here is what's called the negative elbow or the free energy. It has two components. It measures the quality of the reconstruction and it measures how simple, um, your latent space was in terms of staying constrained to the traditionally, uh, a mean vector of zero and a covariance of one. Now, uh, we can generalize this and what we do now is we're going to use a coupled Gaussian

and then we're going to match that coupling with a metric, which is the coupled free energy. And it turns out while the derivations are complicated, in the end, we get out the same metrics that are used traditionally with some additional coefficients. So if we assume the pixels are continuous approximately, then, um, uh, well, let me start actually the latent layer then is measuring the divergence. And that's basically just the trace over the, um, the, the, what's now called the correlation metrics. And then also the differences in the means between the prior distribution and the posterior distribution. So that's one component. And that is a, just a calculation. And then, uh, we also measure the mean square average. And again, this isn't changed. It's just some coefficients that have changed. It's, it turns out we still get the same metric, but now what happens is because we have this heavy tailed distribution as the model, um, where this mean squared average is going to be highly sensitive to what's happening in the tails. And the way we balance that now is that rather than sampling the latent space and the way we balance that we balance directly, we rather sample this independent equals modification to it. And this is the real innovation that's made a big difference now. Um, and again, what happens is as the coupling of the Gaussian goes to infinity, the coupling of the samples we're going to generate is limited to less to be less than one half. So we always guarantee that there's a well-defined mean and variance, uh, uh, from the samples that we're, uh, drawing from. So let's see how this works. Uh, show a couple of pictures of this. So on the left, I have, um, a log linear plot. And then on the right side is the log log plots. And if we start with this light blue curve, that looks like a parabola here, this is the Gaussian and the coupling is 10 to the minus five, which is essentially zero. And the modification doesn't really change that at all. It's still just 10 to the minus five. And again, we see this Gaussian here as a knee on the log log curve and a very sharp decay of the exponential. Then if we go to a Cauchy distribution in this dark, darker blue, we see this very flat tail curve here and you see the longer tail here on the log log plot, but the knee is still in the same place. Now, when we take that independent equals curve, um, we have a much softer decay, excuse me, we have a much faster decay here. Now the coupling is one third. And again, that's less than a half, which is the critical point here. So we took a distribution, the Cauchy that doesn't have a defined mean, the variance is infinite and we modified it into a distribution that's well behaved. And again, so we see that as this sharper curve here on the log log plot. And then finally, we can go all the way to the extreme of a distribution with 10 to the five coupling. There's not necessarily practical use cases for such an extreme distribution, but the point that we can actually still have convergent learning with such an extreme distribution is a unique capability. So we start with this sharp spike here, this very long tail, but both the coupling and the, um, correlation matrix get modified such that the modification is more like a delta function here, showed here in red with the dash. And you can see the, on the right here, the very sharp turn here has to do with the fact that the coupling is so high, but now, um, we have a curve that's more balanced here at the top. And again, this, uh, new curve or distribution has approximately one half as its coupling value. So what does that enable us to do? Um, going back to this example of modeling the Bitcoin returns

in traditional machine learning with the, uh, CVAE, we would be using a Gaussian as the model and it's, it's a mixture. So it's going to get a little bit of the tail, but ultimately it fails to capture these extreme elements of the Bitcoin log returns. Um, so the blue is the actual data. And then this sort of, um, orangey gray color is, um, the model. If we set our coupling to be 0.6, which we measured to be approximately

what that, uh, particular, uh, set of data was characterized by. Now we can see that we do have, uh, samples all the way out into these extreme tails. And so this is the key point here now is that we can, we can now have a way to

model and learn, uh, these extreme distributions. Now to date we'll, we'll make progress on this, but to date the coupling is a hyper parameter, um, that we're setting a priority to the training.

And that, but what is important is that we can then learn that correlation matrix, um, to fit, uh, the data.

Is this still over dispersed on the Bitcoin there?

The model isn't perfect if that's what you're referring to. Um, so yes, it, you know, um, it does look like we're getting, um, some character, some samples that are further out than what the data was.

Is that what you mean? But yeah. And, and then, and then again, just how would this differ than, than an inflated tail? Was it, does it have a different computational requirement or does it just move probability density from a central tendency out to a tail in a way that has a certain, um, I guess, as kurtosis? Yeah. So let me, let me talk about that in two ways. If we stayed with the Gaussian distribution, uh, the problem is that we would end up with this very fat distribution, potentially, like if we really wanted to capture this sample out here, we'd have to have a very fat standard deviation. And then that would mischaracterize the fact that most of the samples are in near the mean. Um, if you go, let's see if I go all the way back to this curve, we often talk about the tail being slower, but another feature of these heavy tail distributions is that around the mean, it's more spiky. So there's a lot of mass near the mean, and then it spreads out. And so what ha so, so it's both of these

characteristics that you want to be able to capture in your model. So that's one aspect. So the Gaussian isn't going to be able to do that. Now there's a second question. Oh, is the, the second point is the following. And maybe the easiest graph to show this is if I go to this graph here, the question might arise, why can't I just use a heavy tail distribution in the latent space, but keep the Shannon entropy metric, right? Well, the problem is what happens is

Dr. Michael Hicks- The Shannon metric is going to be so sensitive to, uh, the uncertainty caused by the tail that it will not have any uns, uh, it will not be sensitive to the scale, which is what you're trying to learn and modify the parameters of in your neural network. Um, and in particular, what happens even when, so I'll go one other thing here in the early work, we have a paper, we're running a new paper on this currently. There's a paper that we published around 2020 on the couple VAE where we use the generalized logarithm, but we sampled from the latent distribution. We didn't really use this part of the function. And what we found there is we were limited to around

a coupling value of 0.5 to one. And as you tried to go to higher values of coupling, the training process just wouldn't converge. Uh, so eventually you'd get, um, such extreme measurements that the training process would just crash and, and you wouldn't have any learning process at all. And so to date, the state of the art is that yes, you can train heavy tail distributions, but typically they're limited to distributions that are of, uh, faster decay than the Cauchy distribution, you know, which is coupling equal to one. And it really doesn't work very reliably unless the coupling is less than 0.5. And I'll, I'll just jump ahead here. What we're showing now is that even in this most extreme case of 10 to the fifth, where you can, I'm not convinced that it has practical value, but it's still converges. I still get a, a neural network with a model. Um, so this is showing now, um, let me, let me just back up a second. So we're, we're doing this test over the, um, celeb a dataset, uh, where we have hundreds of thousands of images of celebrities. Um, and again, the key thing here is that in this latent distribute, I'm going to learn the latent distribution. And after the training, I am going to sample from that latent distribution directly, but during the training process, I'm going to sample from this modified distribution to guarantee that the training process converges. Um, and we show that in fact it does. Um, so against this metric of the coupled free energy, we get continual improvement as we, um, increase the coupling all the way up to 10 to the five and 10 to the five, it's just a flat line here. Now there is a caveat that, um, this is not a one-to-one comparison because we're modifying the model and we're modifying the metric. Um, but the point thing is, is we do show that, you know, the amount of variation we get as we, as we increase this coupling value gets quite low. And, and also just it's, I think it's a brand new result just to be able to learn anything about a distribution so extreme. Um, now.

What is the, the free energy there? The Y axis?

The Y axis that is this coupled free energy. So it's the generalization of the free energy.

And again, the way it is, is that as we go to higher coupling values, it's more sensitive to what's happening in the tail of the distribution. So you can think of it as like a robustness metric.

But at the same time, we've modified what, what samples we're drawing over in order to make sure that this very stringent metric can still converge. Um, now in order to do, uh, we, we have then looked at, um, metrics where we keep the metric constant in order to do the, um, head to head comparison.

And here we see what we would kind of expect is that we get improvements in the accuracy.

And I'll call coupling equal to zero still a accurate measurement. Whereas the positive values of coupling I'll refer to as robust metrics. We do get, um, as we go from 10 to the minus five to 10 to the minus two, we do get improvements in the accuracy and throughout the whole epic. Um, we're getting an accurate model. Um, and that is up to, I guess the, I think the darker one here is, um, 10 to the zero. Oh, actually 10 to the minus one is the darker one. Now, when we go to the Cauchy distribution and beyond the Cauchy distribution, we do see that the accuracy metric with this free energy does show decays in the performance. So you, in this case, the training has kind of us optimized point very early in the epic cycles. And then as you continue to train it, the model actually gets worse with regard to accuracy. Now that's not unexpected because we weren't optimizing for the accuracy.

We were optimizing for something else. Um, and then the other thing, this is labeled incorrectly, but the, um, the last one here is the extreme case of 10 to the five. And it does show that this extreme model has kind of funky behavior and that that's sort of typical of what would have been expected. But again, I emphasize without that use of the independent equals, the whole thing breaks down at more like 10 to the, uh, 10 to the zero here. That was sort of the limit of the prior results. Now we also looked at how, uh, the reconstruction fidelity is for this. Um, and the, on the left, is the original samples in the middle is the, uh, traditional VAE with the coupling equal to zero. And by I, you can maybe make out a little bit of improvement with the coupling value, but the metrics are actually quite impressive. Um, so against a metric known as the multi-scale structural similarity, we see 25% improvement and that stays, uh, consistent across all of the coupling values. So even just a little bit of coupling, getting away from that, the extremes sharpness of the exponential tail, right away, you get performance improvement and that performance improvement stays consistent across all the coupling values that we used. Um, likewise, um, something called the learn perceptual image patch similarity had a 10% improvement. Um, and I guess, okay, so that, that's it. So the conclusion here is that we now have the ability to model the most extreme distributions, and we're particularly, um, interested in how this could be applied to catastrophic risk modeling, number one. Um, but also I, I have a lot of scientific interest, uh, in terms of how it will improve active inference for the following reason. If you're going to measure errors in every single node. So again, just remember the contrast here right now, the way machine learning algorithms are trained is that you have one cost function that's going to, um, be used to modify the parameters of a large network. Um, you can argue that in that situation, uh, the Gaussian and the entropy measurements may be adequate for most use cases. Maybe not if you were trying to model catastrophic risk, but for other use cases, we've seen that it's very powerful technique. This is the engine of AI. But in active inference, because you're measuring errors at every single node, you necessarily have very limited amount of data characterizing those errors. And that's exactly the original use case for the student T distribution was that a low amount of data corresponds to a low degree of freedom. And it's, and that degree of freedom is actually just the inverse of what I'm calling the nonlinear statistical coupling. So another way that these models are grounded in traditional statistics is that, uh, the nonlinear coupling is just the inverse of the degree of freedom. So I think that in the active inference models, um, there's a real necessity for this type of metric because there's a necessity, um, to be able to, uh, model accurately the fact that you're trying to take these error measurements over a limited amount of data. And, and the novelty here is to have both a generalization of the metric in this coupled logarithm and the generalization of the samples you're using with this independent equals, um, and new work that we're making progress on is, uh, the theoretical proofs, um, using information, GMR geometric techniques, you know, that the, this is exactly the correct statistic. Um, and I'll also say like, just for maybe going to future reading here. So I have a preprint that's currently under review that provides the proofs that there's a unique scale called the independent,

um, uh, excuse me, um, that I refer to as the informational scale that is required for the modeling of heavy tail distributions. And then from that, there's only one generalization of the entropy function that gives you the correct measurements. Um, and so the paper is called on the uniqueness of the coupled entropy. And this, a variation of this presentation was presented with a paper at the, um, artificial general intelligence conference 2025. Um, we have a new paper that's, we'll have more details coming out soon. Uh, but the original paper, um, from a few years ago is called coupled VAE. And there we showed improvements in the robustness using these techniques. And then another interesting paper I wrote several years ago was a book chapter, um, called reduced perplexity, which was a play on words.

Um, but it was showing this idea that you can think of entropy as your measurement of uncertainty along the Y axis of a distribution. So with that, I can, uh, thank you all. I really appreciate the opportunity to talk to this community, learn more about what you're doing and looking forward to some nice collaborations together. Great. Yeah. So people watching live can write some questions. I'll ask some, some random things. So these are all central tendency distributions. How do you deal with situations where you just are not mean centered around zero?

Yeah. Okay. Excellent question. Um, so α equal to one is the generalized Pareto distribution, which typically is a one sided distribution. Um, so that could also be the model. Um, and then, um, with Ugar Turcnelli, we are doing, um, some research now in which α is not equal to an integer. And those models have different kinds of characteristics. That's number one. The other thing is that the family itself includes that, um, H of X parameter. So another generalization is what we call the coupled Weibull distribution. So the Weibull distribution is another popular one sided, uh, distribution from the exponential family. And then you can generalize that with this coupled exponential. So we're also looking at that as an interesting case. Yeah. Okay. Okay. Now I have, uh, um, I'm happy to take questions from the audience,

but I have a question for you, Dan, in your work on active inference, have you noticed this feature that because you're trying to learn at every node of a network that you have difficulties with the limited amount of data that each node, um, experiences. So there's nothing about these models that intrinsically has low or high amounts of data. You could train a massive data set on a single parameter or, or have a situation where there's very little data. Um, not always just a low number of data points per parameter to learn like the sort of large, uh, P small N issue, but especially getting into thinking about causal effects of action, having world models or just having generative models that have a lot of conditional aspects makes it such that you will not have data for a large number of conditional aspects, which sort of forces factorizing or simplifying the problem back down to simpler, uh, lower dimensional sub problems. And so the challenge would be, well, if those lower dimensional sub problems could be solved, but then you have even a lot of combinatorics of which smaller dimensional sub problems to tackle. And again, if you knew the structure of the problem, then that is very tractable structure learning, but then in a situation where you might be dealing, um, with just different kinds of data streams and not necessarily have a well-structured world model, then even that

structure learning problems becomes very challenging.

Mm-hmm. Mm-hmm.

Yeah. Uh, well, and yeah, you actually bring up another interesting use case, which is the curse of dimensionality. So as you go to higher dimensions, this issue of needing to use the heavy tails to, in order to make sure that your models are more robust and more accurate becomes more prominent.

Yeah.

On the heavy tails, I wondered, have you thought about any of the physical connections with the entropy?

Yeah.

So there's obviously more generalized and informational entropies than necessarily happening in a cup of water. But how do these different tail distributions, for example, connect with temperature distributions and thermal movements of particles? Do they all follow one type of tail distribution or do different materials or different physical situations follow one of these differently?

Yeah. This is actually the heart of what scientists in the field of complex systems work very actively on.

And, you know, thermodynamics grew out of modeling equilibrium type things. So temperature, for instance, is a measurement that basically has a requirement that you're in equilibrium. But of course, we're very interested in being able to characterize non-equilibrium systems

or what can be called non-equilibrium steady state systems, where it's not in equilibrium, but it does, but the state does persist over time. And there's a lot of work in that area. This is a big part of what motivated a field called non-extensive statistical mechanics that was originated by Constantino Salas.

And there's been proposals to generalize thermodynamics using modified entropy functions.

Now, I would argue, but it's not fully proved that mostly those attempts have not been very successful.

They've had modest success, but they've not really come up with a complete model.

And in this new paper, I have a section showing that you do get a rigorous definition of the temperature as being equal to this scale parameter. And again, this is another thing that I think makes both the way I'm characterizing the distributions and this particular entropy function unique, because I show that you can, that basically that scale parameter is your generalized temperature.

And the interesting thing about that calculation is that

that in order to get that, what came out of the mathematics is that the free energy function that underlies thermodynamics, it's that function that gets generalized and has an additional nonlinear term in it. And so that's where the generalization comes. And then the other thing is, I showed that

there can be a complementary model where I can define the temperature to be the scale raised to a power related to that coupling term. Now, my definition of temperature is more complex

and involves the non-linearity. But in that case, then I can have no modifications to the free energy equation. So there's also sort of this interesting duality in terms of how you build

the models. But I think with that result, there's a strong case and it would be very worthwhile for people to look at whether the full

laws of thermodynamics can be generalized using the coupled entropy.

So back to the data, it's interesting that you highlight learning from little data, whereas almost

one of the key advantages is that you will not deflate the variance when you have a lot of data.

Because the main issue is not having a wide estimator on sparse data. That's free.

The hard thing is when you have a lot of samples not to just converge back to a training data set. And that's why

variational methods and all other kinds of entropy-regularized methods kind of are out there.

Because you don't want a collapse simply to recapitulating the training data or even narrower.

Yeah, yeah. Well, I yeah, you know, it was, it's funny how just in the process of working on these things,

you learn a different connection. So I have, my background is in electrical engineering. And as

part of that, I've studied both mathematics and physics quite a bit. So I had not, I would say a

strong background, but certainly a background in thermodynamic processes. At the same time, as we

were developing the variational autoencoder, we were learning about the importance of the evidence

lower bound. And it's a basic way that it can be decomposed into a measure of accuracy in your

generative process. And then also a measure of the regularization of making sure that you using a

simple model. And that it's the balance between these two things that's important. And ultimately,

what gives you a good model. So it's not just accuracy, but it's accuracy with the simplest model

possible. Now, it was only through, it was only through reading Carl Fristen's work on the active

inference that I had this, gained this understanding that the evidence lower bound function is exactly the

free energy. And so there is a direct connection between the thermodynamics of complex systems and the

training of artificial general intelligence and generative models. And so I, as we as a team have

been reading, um, par and Pristin's new book on active inference, it's been fascinating to see

just how general the concept is. And, you know, again, another thing that's exciting for us is that we've

mostly focused on variational inference. And six months ago, I was sort of expecting that as we migrated to

decision making, and wrote, you know, kind of robotic control, that that would require a whole other set

of algorithms and a whole another set of generalizations that we would have to work on. But now, of course,

active inference has integrated all of these techniques. You know, so, so now I have a better understanding that

when it comes to action, you can think of your utility function as just a prior distribution.

And then what action you want to take, you can think of as an expected

! Um, minimization of the free energy. So it's a beautiful representation. And we're really excited

about working on these different kinds of problems.

Let's sort of foreshadow and head off on one area that we've talked a little bit about, which is when

we're dealing with causal models and structured learning, we're in a little bit more of an expanded

setting than just like a financial returns model. So the financial returns, we're dealing with

something that might be tightly clustered, or widely dispersed, or have an average trend, or be zero

centered on a given fixed, you know, 24 hours percentage change. But yeah, and as you get, there's

more and more ways to be extreme. And there's fewer and fewer ways to be essentially conformist.

Yeah.

Now when we open the door to cognitive systems and multi-agent systems, and we're thinking about

complex systems in general. So there's more and more and more and more ways that couplings amongst actual subunits can interact. So what kinds of macro properties or approaches might help us bound where we have uncertainty, not just over kurtosis properties within a distribution, but where we actually have a structural uncertainty about like geospatial causal models?

So I'll answer this a little bit indirectly. But I think something fascinating that comes out of this is another core principle in the active inference model is the idea of a Markov blanket, where you separate your internal model from the external model. And then a market blanket is where you declare independence and sort of the, and another way it can also be thought of, like one way you can think of it as the Markov blanket between the past, the now and the future, right?

Well, that's certainly foundational. But one of the things that comes out of this coupled algebra is the ability to relax that assumption of independence. Because if I apply now rather than a Markov blanket, what I might refer to as a coupled Markov blanket, I am going to have independence in terms of correlation matrices. But I'm going to have, be able to maintain a model in which there's actually coupling across the blanket. So the coupling parameter is now going to have, be a way of storing long-term correlation.

And I have done a little bit of work on this in papers that either refer to coupled Markov processes or coupled Bayesian processes. And, and, and at the heart of each of these, again, the Bayesian calculation is similar. One of the basic assumptions in Bayesian analysis is that the prior distribution gets multiplied by the likelihood. And then that gives you the posterior distribution. Well, that multiplication is assuming that the prior and the likelihood are independent of each other. But with the coupled algebra, you can relax them. And you, you can have a nonlinear coupling between your prior state and your posterior state. So for instance, there's a scientist topsoy who's used that model in models of cognition.

And, and the way I think about his models is sort of like how, how strongly you care about your prior. And so, you know, you, you might be a person that's sort of tied to a particular set of beliefs, right? And no amount of data is going to change those beliefs. And you can model that with this coupled algebra and then vice versa, you might be someone who doesn't have, you know, you're very loose in your priors and you're very loose in your beliefs and you kind of then easily swayed by whatever data comes in next. And these kinds of differentials are some of the more interesting models that can be created with this mathematics.

Now, I don't think that answered your question directly, but it, but the point is that there's much more subtle kinds of causal models that can be created using these techniques.

Well, I like the idea of the Markov blanket as the declaration of independence, the fun idea, and definitely the attention to past, present, future, or just the Markovian property of time with the past influencing the future through the present with this Bayesian moment of the intersection of how you couple the prior and the likelihood. And then also in that, it makes me think about, yes, we could talk about all of the interactions among subunits and there's combinatorics of structural possibilities for the system of interest. However, the variational modeling agents may have a much simpler model that they would have more defined properties of. So the estimators model properties, the map, do not need to recapitulate actual structural adjacencies or aspects of the world.

And intelligence is about that kind of Bayesian information criteria or similar driven approach to ride on that edge with a dual imperative of accurate and small.

Yeah. Ultimately, it is part of the goal is to get simpler models. I agree with that. And one way you can picture that is that if you have these heavy tail issues, one way you can get a model still using the Gaussian is to have a mixture of many Gaussians with different variances, right? And so in a sense, what the heavy tail is giving you is the ability to have one distribution that can characterize a large mixture model of many Gaussian distributions. So it does... So I do think that as you incorporate the heavy tail into the foundation of these models, you'll also be able to simplify how many mixtures you need in order to get a good characterization.

So I think that's a good question.

I think that's a good question.

Just last comment or note here. What kind of software tooling or what repos or which of the papers should people explore for if they wanted to get hands-on?

Yeah. Right. So the two papers on the machine learning have references to our non-linear statistical coupling and CVAE repositories on GitHub. And we... The NSC library just uses a basic Python tools. And then we originally did the work in... I'm sorry, TensorFlow. And then we migrated more recently to using PyTorch.

Cool. So I hope that people walk through this and explore the tooling. And if there's anything else, there's anything else, please feel free to add it. Otherwise, this has been very interesting, connects some of these basic statistical principles, basic but advanced statistical principles, to some of the real situations that embedded systems and intelligence systems are finding with, you know, too little, too much data, events that have more variability than we have seen in training data. These are issues that beset all. So it's awesome to see how you're researching and applying that.

Yeah. Super. Thank you very much for the invitation, Daniel.

Okay. Farewell. Bye.

Take care.

Bye.